Foundation II: Shared Manifestation Shared Structure, Public Calibration and Different Histories

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Abstract

Several readers of one process can have different accessible algebras while sharing a world. Their common structure is the intersection of their stable hulls. It is the largest algebra available to all of them, inherits the common modular flow and, with a supplied positive translation structure and a unique vacuum, contains a proper half-sided modular inclusion whenever it is non-scalar. No foreground overlap or commutation of noise channels is required. For compact symmetry-selected hulls the common algebra is fixed by the group generated by the readers' stabilizers; under the minimal-action Galois conditions their jointly generated algebra is fixed by the intersection of those stabilizers. The SU(2) level-one model realizes a shared Virasoro sector between two different current sectors, even with disjoint foreground seeds.

Agreement does not certify the truth of a shared description. A common symmetry change of instruments and preparations is indistinguishable, in every finite adaptive record law, from the corresponding change of actual state. We derive the effective groups of these errors and show that their kernels differ on individual and shared content. Conversely, a non-scalar translation-stable shared algebra contains a finite positive-energy pulse. With preparation and readout permission, a known vacuum baseline turns it into an energy-orientation reference with an explicit finite-record error bound. Under the additional source regularity and modular-consistency conditions of Consciousness II, the pulse also supplies a spectral witness for a common sourced geometric response. It does not fix a metric without boundary or initial data. For a finite family of common frame errors, legal reference blocks identify a public target exactly when indistinguishable reference laws have the same target value. The resulting finite-record bound permits residual private frame differences; an explicit reference protocol preserves different old histories. A two-observer calibration model measures common error and fault transfer. Independent-source compatibility is treated separately. Shared carrier, shared state, agreeing records and correct content are distinct requirements.

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1 Introduction

Foundation I asks how a source, its noise and a retained seed specify an accessible algebra [1]. This paper asks what several readers of that source share. The question has two parts. A common carrier must be constructed from the readers' access. A common description must also survive errors that all readers can make together. Agreement alone addresses neither part completely.

The main setting is one ambient process with different seeds and accessible hulls. Their intersection derives the shared algebra instead of assuming a candidate carrier. Its invariance needs no commutation of the noise channels. If a positive translation structure is supplied, a non-scalar intersection inherits a Cut on its own vacuum-cyclic space. Cuts sit inside that shared algebra; the algebra itself is not asserted to be a Cut of the ambient algebra. The translations remain physical inputs, as in Foundation I and Consciousness I [1, 5].

Symmetry makes the distinction between common and joint access precise. For hulls fixed by subgroups, the shared algebra is fixed by the group those subgroups generate. The joint algebra obeys a different law under the compact Galois hypotheses [3]. Two independent current seeds in the SU(2) level-one net have no foreground overlap yet share a Virasoro sector. The algebraic sharing mechanism is therefore stronger than foreground matching. Foundation III provides a noncommuting, repeated-noise realization of the same fixed intersection [2]; the present paper uses that result without repeating its convergence proof.

Readers can then agree and still share a displaced standard. Reference-frame restrictions and relational encodings are established subjects in quantum information [4]. Here an adaptive-protocol covariance identity identifies a class of common errors that no agreement statistic can expose. The group acting faithfully on individual content can be larger than the group acting faithfully on shared content. In the current-net model a common energy sector is insensitive to reversals of current orientation. This is an operational distinction, not a claim that a symmetry-invariant reading is automatically true.

A shared algebra can also contain a way to test a standard. Consciousness II's bounded-preparation construction gives a finite positive-energy pulse whenever the shared algebra is non-scalar and translation-stable [6]. If the pulse and vacuum baseline can actually be used at a common energy port, the mean difference determines an otherwise unknown energy-gain sign. The preparation changes the physical state, so it can also provide the source witness required to identify a geometric response, subject to the separate regularity and consistency conditions. This connects selected access to a testable world rather than identifying geometry from consensus alone.

The source need not have been described by one state in every problem. Independent source states, projections and memory kernels have a different compatibility question. Appendices A–D retain that theorem and its nonempty examples. It cannot be made unconditional by substituting the one-process construction for its independent states. Likewise, the two-observer simulation of Section 5 is a finite-resource calibration experiment, not a finite reconstruction of a Type III algebra.

Quantum Darwinism and spectrum broadcast structures study information held redundantly in environmental fragments [20, 21, 22]. An intersection of accessible algebras is not a redundancy certificate, and two carriers can encode the same bit even when their literal algebraic intersection is scalar. The paper keeps these distinctions while connecting the source-selection problem of HAFF to the reference and self-calibration problems of T-DOME [9, 10, 12, 13].

Section 2 derives shared structure and its symmetry laws. Section 3 identifies common frame ambiguities. Section 4 constructs the internal reference and its conditional geometric

test. Section 5 gives the calibration simulation, and Section 6 returns the results to the full theory.

2 One process, several accessible structures

Let M be a von Neumann algebra with a cyclic and separating vector Ω and faithful normal reference state $\omega(x) = \langle \Omega, x\Omega \rangle$. A family $\mathfrak{E}$ of normal unital completely positive maps preserves ω. Its members need not commute. Include the full reference modular group and put

$$\Gamma_2 = \{\sigma_s^\omega : s \in \mathbb{R}\} \cup \mathfrak{E}.$$

When a positive translation group $U(a) = e^{iaP}$ is supplied, with $P \geq 0$, $U(a)\Omega = \Omega$ and $U(a)MU(a)^* \subseteq M$ for $a \geq 0$, also put

$$\Gamma_3 = \Gamma_2 \cup \{\mathrm{Ad}\, U(a) : a \geq 0\}.$$

For either family, a subalgebra is admissible when every map in the family takes it into itself. A reader retains a seed $S_j \subseteq M$ and has the ideal accessible hull $B_j = \mathrm{Hull}_\Gamma(S_j)$: the intersection of all admissible von Neumann subalgebras containing S_j [1]. This closure describes an operator structure. It is not a finite-resource reconstruction algorithm or a licence to implement every unitary in the hull. The seed, noise, control permissions and reference state are physical inputs.

Theorem 2.1 (The shared structure of one process)**.** *For any family of readers with the same M, ω and Γ, set $B_\cap = \bigcap_j B_j$.*

1. *$B_\cap$ is the largest admissible von Neumann subalgebra contained in every B_j. It contains* $\mathrm{Hull}_\Gamma(T)$ *for every $T \subseteq B_\cap$, and carries a unique normal ω-preserving conditional expectation. Its modular group in $\overline{B_\cap \Omega}$ is $\sigma^\omega|_{B_\cap}$.*

2. *For every normal actual state ψ of M, the restrictions $\psi|_{B_j}$ agree on $B_\cap$. This says nothing about equality of the readers' joint records or about the accuracy of their decoding.*

3. *For Γ_3, suppose U is the translation group of an ambient half-sided modular inclusion and* $\ker P = \mathbb{C}\Omega$*. If $B_\cap \neq \mathbb{C}1$, then $U(a)B_\cap U(a)^* \subset B_\cap$ for $a > 0$ is a proper half-sided modular inclusion on $\overline{B_\cap \Omega}$. The recovered translations are the restricted ambient ones. If the inclusion is standard, Consciousness I's Cut–geometry correspondence applies there.*

Proof. An intersection of von Neumann algebras is a von Neumann algebra. If x is in every B_j, then $\gamma(x)$ is in every B_j for each $\gamma \in \Gamma$. For the modular maps, including both s and $-s$ gives equality rather than only inclusion. The intersection is therefore admissible, and every admissible subalgebra contained in all the B_j lies in it. If $T \subseteq B_\cap$, the intersection is itself a competitor in the definition of its hull. Modular invariance gives the expectation and restricted modular group by Takesaki's theorem [17, 16]. Restrictions of one state agree on an intersection. Finally $B_\cap$ inherits forward translation invariance; Foundation I's Cut-inheritance theorem supplies (3), including properness from the unique invariant vector hypothesis [1, section on Cuts in the hull]. □

This route needs neither a candidate shared carrier nor commuting projected memory kernels. It also does not require the foreground seeds themselves to overlap. It addresses readers of one process. Independently specified source states can still disagree on an overlap; their compatibility is treated in Appendices A–C. The reference ω determines the closure and the Cut. The actual state ψ may be different, even nonstationary. Its restrictions agree because they are restrictions of one state, not because the reference and actual states have been identified.

2.1 Shared, individual and jointly available algebras

Let a compact group G act continuously on M, preserving ω, and let K_j be closed subgroups such that $B_j = M^{K_j}$. Write

$$K_\cup = \overline{\langle \bigcup_j K_j \rangle}, \qquad K_\cap = \bigcap_j K_j.$$

The fixed-algebra identity

$$B_\cap = M^{K_\cup} \tag{1}$$

holds without a minimal-action assumption: an element fixed by every K_j is fixed by their generated group and, by continuity, its closure. If the action is minimal on a factor with separable predual, the compact-group Galois correspondence also gives

$$\bigvee_j B_j = M^{K_\cap}. \tag{2}$$

Here minimal means faithful with $(M^G)' \cap M = \mathbb{C}1$ [3]. Indeed the join is an intermediate algebra M^L. Containment of every M^{K_j} forces $L \subseteq K_\cap$, whereas the join lies in $M^{K_\cap}$, forcing the opposite inclusion. If $M^{K_\cap}$ is also Γ-admissible, the hull of the joint seed equals that join. Admissibility of a join is not automatic for arbitrary UCP maps, which need not preserve products of elements from different subalgebras.

Example 2.2 (Shared geometry without shared foreground)**.** In the SU(2) level-one current net let $M = \mathcal{A}(0,\infty)$, with its vacuum, modular dilations and forward translations. With no additional noise, the bounded current seeds $\frac{2}{\pi}\arctan J_{e_3}(f)$ and $\frac{2}{\pi}\arctan J_{e_1}(f)$ have hulls M^{H_z} and M^{H_x}, where H_z and H_x are the axial rotation groups. These seed-to-hull statements and the fixed subnet are established in Foundation I [1]. Here f is nonzero, real and smooth with compact support in the positive half-line. The two groups generate SO(3), so

$$B_\cap = M^{SO(3)} = \mathrm{Vir}_1(0,\infty), \qquad B_1 \vee B_2 = M.$$

Each individual hull is a current algebra; their common algebra is Virasoro; their joint algebra is the full current net. They are three different objects. The two seed lines have zero foreground intersection, yet their hulls share a standard Cut. Foreground overlap is therefore not necessary for sharing. The supplied vacuum net and translations remain inputs, not outputs of the sharing argument.

Nontriviality is not universal. In $M_2 \otimes M_2$ with the trace, trivial modular flow and no noise, the seed algebras $W^*(Z \otimes 1)$ and $W^*(1 \otimes Z)$ intersect only in the scalars. The two registers can nevertheless encode the same classical bit with high agreement. A shared subalgebra on one ambient carrier and matching content on different carriers are distinct ways of sharing; neither defines the other.

2.2 A dynamical realization without commuting noises

For a finite family $K_1, \ldots, K_m$ let Q_j be the Haar expectations and apply $D_n = (1-\theta_n)\mathrm{id} + \theta_n Q_{i_n}$, with $0 < \theta_n \leq 1$. Foundation III proves

$$\sum_{n:i_n=j} \theta_n = \infty \quad \text{for each } j \quad \Longrightarrow \quad D_n \cdots D_1(x) \longrightarrow Q_{K_\cup}(x) \quad \text{strong*}. \tag{3}$$

Thus $B_\cap$ can be the algebra retained by repeated noise, not only an intersection written after the fact [2]. There is no bounded-gap or commutation condition. Record-dependent schedules are covered when their realized sequence meets the exposure condition, but intervening measurement and control channels must be included separately. The finite-family restriction is essential; Foundation III supplies a countable family for which every noise is visited indefinitely but convergence fails. This constructs persistence for supplied noise actions. It does not derive why the source has those actions, or turn the ideal limit into a finite-time resource guarantee.

3 Common errors that agreement cannot expose

A reader's standard assigns record labels to instruments, preparations and decoders. Correlating standards is a reference-frame problem, not the equality of two restricted states [4]. Let α_g be a normal automorphism of M commuting with every uncontrolled evolution map used in a protocol. Each instrument branch $\mathcal{I}^y$ is a normal completely positive map, and its sum over outcomes is unital. Choices of later branches and preparations may depend on earlier labels and on recorded private randomness, with the same prescribed conditional coin law in the two configurations being compared.

Proposition 3.1 (An indistinguishable common change of standard)**.** *For any finite adaptive protocol, replacing every controlled operation by $\alpha_g \mathcal{I}^y \alpha_g^{-1}$ in the actual state ψ gives the same joint record law as the original protocol in $\psi \circ \alpha_g$. If $x = x^* \in M$, $\alpha_g(x) = -x$ and $\psi(x) \neq 0$, a binary report about the sign of $\psi(x)$ has correctness probabilities adding to one in these two indistinguishable configurations. At least one of them has correctness at most $1/2$, regardless of the agreement rate or the number of readers.*

Proof. For a fixed record string the probability is a state applied to the ordered composition of instrument branches and intervening evolution maps, evaluated at 1. Conjugating the controlled branches and commuting α_g through the uncontrolled maps leaves α_g on the outside and $\alpha_g^{-1}(1) = 1$ on the inside. This changes ψ to $\psi \circ \alpha_g$. Adaptivity changes the branch selected by the fixed string, not this identity. Averaging over private coins preserves it. The truth changes sign between the two configurations while the report law does not. A binary report is correct in exactly one of them, proving the sum and the bound. □

The ambiguity concerns two physical configurations: a displaced standard and a displaced state. It is not the assertion that the same state has two different true values. A calibrated reference left unchanged by the protocol can break the symmetry; conjugating that reference as well would remove the very information that makes it a reference. This result classifies a source of common error, not every statistical mistake. Two independent error bits with error probability e still agree wrongly with probability e^2.

3.1 The effective group of a common frame error

Let G be a compact group of allowed frame transformations, and let $H \leq G$ be its closed subgroup preserving the uncontrolled world laws. Suppose the individual algebras are $B_j = M^{K_j}$. We first consider errors that preserve each labelled reader's algebra rather than exchange the readers. Define

$$N = H \cap \bigcap_j N_G(K_j), \qquad L = N \cap \bigcap_j K_j, \qquad L_\cap = N \cap K_\cup.$$

Under the minimal compact-action hypotheses used in (2), the pointwise stabilizer of M^K inside G is exactly K [3, Remark 3.17].

Proposition 3.2 (Joint and shared errors have different kernels)**.** *In this regime the transformations in H preserving every B_j are exactly N. Their faithful action on the tuple of individual algebras, or on their jointly generated algebra, is N/L. Their faithful action on the shared algebra $B_\cap = M^{K_\cup}$ is $N/L_\cap$. In particular $L \subseteq L_\cap$, so a nontrivial error on individual content can be invisible on all shared content. For one reader the effective group is*

$$N_H(K)/(K \cap H),$$

where $N_H(K) := H \cap N_G(K)$ even when K is not contained in H. This reduces to $N_H(K)/K$ only when $K \subseteq H$. For one reader the observables unchanged by every such error form $M^{\overline{\langle K, N_H(K)\rangle}}$.

Proof. $\alpha_g(M^K) = M^{gKg^{-1}}$, so injectivity of the fixed-subgroup correspondence makes preservation of M^K equivalent to normalization of K. The pointwise kernel on each B_j is $N \cap K_j$; intersecting gives L. Fixing every individual algebra pointwise is equivalent to fixing its join pointwise. Every element of N also normalizes $K_\cup$, and the pointwise kernel on $M^{K_\cup}$ is $N \cap K_\cup = L_\cap$. Both are kernels of continuous group actions, hence closed normal subgroups of N, even if K_j is not contained in H. Also $K_\cap \subseteq K_\cup$. The invariant-algebra statement follows by imposing invariance under K and $N_H(K)$ together. □

Without the Galois hypothesis one can still restrict normalizing transformations to the reader's algebra, but the effective group is the group divided by its actual pointwise kernel. That kernel need not be $K \cap H$. The invariant algebra in Proposition 3.2 is immune to the specified frame errors. Its readings can still have statistical error or be decoded incorrectly; immunity is not correctness.

Example 3.3 (Energy is shared while a current can be reversed)**.** In the single-axis SU(2) level-one noise model the world-law symmetries inside SO(3) form the axial normalizer $O(2)_z$, while a current reader has $K = H_z \cong SO(2)$. The residual frame group is $\mathbb{Z}_2$: its nontrivial element reverses J_z but fixes the stress tensor. The immune algebra is the $O(2)_z$ orbifold, not the full current algebra [1]. For the two perpendicular current readers of Example 2.2, with no extra noise, N is the four-element group of identity and the three half-turns about the coordinate axes. Its action on the pair of private algebras is faithful, whereas it acts trivially on their shared Virasoro algebra. The energy sector is unchanged while either current orientation can be wrong. This gives an explicit nontrivial N/L with trivial $N/L_\cap$.

4 A reference made inside the shared structure

Agreement cannot break a symmetry of the whole protocol. A preparation with a known baseline can. The distinction matters for the connection from algebra selection to geometry: a changed reading is not a changed material source, whereas an implemented excitation is.

Proposition 4.1 (Shared positive-energy preparation and orientation)**.** *Suppose the readers' hulls have the common representation and Γ_3 translation closure of Theorem 2.1, with* $\ker P = \mathbb{C}\Omega$*. If $B_\cap$ is non-scalar, it contains a bounded self-adjoint non-scalar operator B obtained by smoothing a translation orbit, and a unitary $W_\theta = e^{i\theta B}$ with*

$$0 < E_\theta = \langle W_\theta\Omega, PW_\theta\Omega\rangle < \infty \quad \textit{for } 0 < |\theta|\|B\| < \pi.$$

All energy moments of this state are finite. If this unitary is an allowed preparation and the same energy port can be used on the known vacuum and on freshly prepared states, suppose its output is

$$Y = \epsilon g\, p + b + \nu, \qquad \epsilon \in \{-1, 1\}, \quad g > 0,$$

where p is the spectral energy outcome and the noise has the same finite mean in both uses. The difference of population means is $\epsilon g E_\theta$, so these record laws identify the sign ϵ.

Proof. The intersection inherits separation and the forward translation orbit of every one of its operators. Consciousness II's bounded-preparation proposition therefore applies [6, section on a bounded accessible preparation]: for a narrow real nonnegative $f \in C_c^\infty(0, \tau)$ with $\int f = 1$ and a bounded non-scalar $C = C^* \in B_\cap$, set $B = \int f(t) U(t) C U(t)^* dt$. As the support of f shrinks to t_0, $B\Omega \to U(t_0)C\Omega \notin \mathbb{C}\Omega$ by separation, so a sufficiently narrow f keeps B non-scalar. Its bounded iterated commutators with P then give finite energy moments of $W_\theta\Omega$. Separation and the unique ground vector give strictly positive energy for non-scalar W_θ. The vacuum has $p = 0$, so subtracting its output mean removes both the unknown offset and the common noise mean. The remaining sign is ϵ. □

P is the common translation generator of the model. In a null Cut it is null momentum, not automatically the time-translation Hamiltonian of a higher-dimensional source. Positivity alone does not supply this calibration: in a finite two-level system $P = \mathrm{diag}(0, 1)$ satisfies $XPX = 1 - P$. The reversal fails to preserve the known vacuum baseline, which the preceding protocol requires. Reading the baseline and implementing the excitation are real permissions, not consequences of algebra membership.

Corollary 4.2 (A finite-record orientation certificate)**.** *Assume independent fresh samples and additive noise independent of the energy outcome, with variance σ_ν^2 common to the two uses. From n vacuum reads and n prepared reads, the sign of their sample-mean difference has error probability at most*

$$\min\left\{1, \frac{g^2 \operatorname{Var}_\theta(P) + 2\sigma_\nu^2}{n g^2 E_\theta^2}\right\}. \tag{4}$$

This protocol uses $2n$ reads and n fresh preparations, whose prepared subsystem energy is nE_θ. Switching, detection, reference preparation and reset costs are separate. A uniform prescribed-confidence budget needs bounds on the gain, noise and energy variance.

Proof. The mean difference has mean $\epsilon g E_\theta$ and variance $(g^2 \operatorname{Var}_\theta(P) + 2\sigma_\nu^2)/n$. A wrong sign requires a deviation at least gE_θ; Chebyshev's inequality gives the bound. □

This reference breaks energy-orientation ambiguity but not every internal frame symmetry. A group commuting with P can still reverse a charge or current. Thus a common positive pulse calibrates one part of the world without supplying a universal external standard.

4.1 Public correctness without equal private histories

Calibrating a public target need not identify the whole frame. Let F be a finite set of possible common frame errors. A legal reference block comprises specified preparations, their source changes, instrument uses and recorded coins. It has a finite record alphabet and known law $p_f(z)$ under $f \in F$. These laws concern actual reference operations that are held fixed when comparing frames. They are not inferred from the agreement of the readers. Let $t(f)$ be a finite-valued public target.

Proposition 4.3 (Target-specific reference criterion)**.** *Repeated independent fresh reference blocks permit uniformly consistent estimation of $t(f)$ if and only if*

$$p_f = p_{f'} \implies t(f) = t(f').$$

When the criterion holds, put

$$r = \max_{t(f)\neq t(f')} \sum_z \sqrt{p_f(z)p_{f'}(z)} < 1.$$

For n blocks, a maximum-likelihood estimate $\hat{f}$, followed by the report $t(\hat{f})$, has target error at most

$$\min\{1, (|F|-1)r^n\}.$$

If no two hypotheses have different targets the error is zero. A single decoded public report transmitted without corruption makes every reader correct on the same event, irrespective of their private histories.

Proof. If two identical laws have different targets, their complete n-block laws are identical for every n. The probabilities of reporting their two different correct values sum to at most one, so one error is at least $1/2$. Conversely, under true f, selection of a wrong-target candidate f' requires its likelihood to be at least that of f. Markov's inequality applied to the square root likelihood ratio bounds this event by $\left(\sum_z \sqrt{p_f(z)p_{f'}(z)}\right)^n$. Zero-probability records are treated on the support of p_f. The union bound over wrong-target candidates gives the result. Distinct probability laws have affinity below one by Cauchy–Schwarz. Finiteness makes the maximum strictly below one. □

The testing inequality is standard; the structural consequence here is that public calibration is determined by the target's indistinguishability classes. A full frame reconstruction would require separation of every distinct frame. When frame symmetries are represented by a group, unremoved transformations may still act on private currents while fixing the public energy target. Identifying a reference law is also weaker than implementing the inverse frame transformation: reporting and control are separate permissions.

There is a concrete finite physical witness. Let a reference qubit have $H_R = \Delta|1\rangle\langle 1|, \Delta > 0$. Legal preparations $|0\rangle, |1\rangle$ and an energy pointer supply the known baseline and excited input. Suppose the pointer's two values are reversed by an unknown common sign $f = \pm 1$ and its binary read flips with probability $e < 1/2$. Reading the excited input gives Bernoulli laws with affinity $2\sqrt{e(1-e)}$, so n reads identify the common sign with error at most $[2\sqrt{e(1-e)}]^n$. An ideal controlled pointer interaction realizes the unflipped measurement; the read noise is a supplied physical channel. The source preparation uses $n\Delta$ in excited-subsystem energy. The pointer preparation, coupling, elapsed time, storage and disposal of spent systems have their own costs. Energy supplied is not net heat.

Let D contain all old private memory registers and their old references, in an arbitrary state ρ_D, with different histories at different readers. Prepare the reference blocks independently of D and apply the calibration instrument only to those blocks. The final state is exactly $\rho_D \otimes \rho_{\text{calibration}}$; all old correlations in D remain intact, even conditional on the calibration outcome. More generally, for initially correlated active and old registers, a trace-preserving map acting only on the active register preserves the old marginal nonselectively, whereas a selected new record can reweight the old history. This is the distinction needed by the retained-history return in Foundation III [2]. The tensor architecture is an existence witness, rather than a factorization theorem for Type III algebras.

The reference criterion fails if the excitation and its label are conjugated along with the faulty instruments: Proposition 3.1 then applies again. It also fails for a destroyed pointer with $e = 1/2$. A shared algebra supplies possible operators; it supplies neither an independently trusted reference nor the preparation and communication operations used here. For a reference block costing resource vector $\mathbf{c}$, n blocks cost $n\mathbf{c}$, in addition to the retained private storage and public communication. Those requirements make the claimed public correctness falsifiable.

4.2 From a shared preparation to a geometric test

Suppose an internal compact group G fixes the bounded functions of the required stress-tensor smearings. Every selected algebra of the form M^{K_j}, $K_j \leq G$, then contains those readings, as does $B_{\cap}$. This is a direct condition for a common stress sector; it is not a theorem about every non-scalar hull. Readers of separate decoupled fields can have disjoint stress sectors.

Proposition 4.4 (A shared test of the sourced geometric response)**.** *In Consciousness II's conformal Minkowski setting, suppose a shared positive-energy pulse is implemented, either the bounded pulse of Proposition 4.1 or an explicit scalar preparation of that paper. Its translation generator is the physical positive Hamiltonian, and its vacuum-subtracted source has an integrable energy density with integral $E_\theta > 0$. Suppose the source and response belong to the smooth tempered Radon Fourier class, and the full centre, open-frame and modular-consistency hypotheses of that paper hold. If every reader has the required stress and boundary observables, uses the same actual state and normalization, and applies the same geometric readout law, the pulse has spectral weight away from zero and identifies their common canonical response up to the law's normalization. They obtain the same matter-sourced linearized Einstein tensor. A unique metric requires additional common initial or boundary data and gauge choices.*

Proof. If a Radon Fourier measure were supported only at zero, the source would be constant. Integrability of its energy density on the infinite spatial slice would force that density to vanish, contrary to $E_\theta > 0$. The spectral identification theorem and common sourced response of Consciousness II now apply [6]. A homogeneous gravitational wave in dimension at least four can have zero linearized Einstein tensor and nonzero linearized curvature, so the source alone cannot fix the metric. □

Finite energy does not itself imply the Radon regularity used here, and the finite-amplitude pulse need not have a nonzero first-order stress tangent at zero amplitude. Nor does the chiral SU(2) model establish all the higher-dimensional hypotheses. The bridge states exactly which source and readout data must be common. A unitary used for calibration physically changes the state; merely choosing a different label or algebra does not. The implemented source and its costs are the link to HAFF-D [10]; bounded source feedback and persistent access are treated in Foundation III [2], rather than inferred from agreement.

Example 4.5 (Shared hulls and a physical time generator)**.** Take two independent massless conformal complex scalar fields in $3+1$ dimensions, on one vacuum Fock space, and let $M = \mathcal{A}(V_+) = \mathcal{A}_1(V_+)\bar{\otimes}\mathcal{A}_2(V_+)$. The future-cone vacuum algebra is standard and its modular group is dilation [8, Theorem 2.3(ii)]; the free massless-field modular structure was studied in [7]. Write $\mathcal{N}_j(O) = \mathcal{A}_j(O)^{U(1)}$ and give the two readers the selected nets $B_1(O) = \mathcal{N}_1(O)\bar{\otimes}\mathcal{A}_2(O)$, $B_2(O) = \mathcal{A}_1(O)\bar{\otimes}\mathcal{N}_2(O)$. For $O = V_+$ their shared algebra is $\mathcal{N}_1(V_+)\bar{\otimes}\mathcal{N}_2(V_+)$. The two vacuum-preserving Haar expectations commute, proving this intersection directly. Specify the noise family to be $\mathcal{E} = \{Q_1 \otimes \mathrm{id}, \mathrm{id} \otimes Q_2\}$, where Q_j is Haar projection on field j. Both maps preserve each $B_i(V_+)$: one acts identically on its fixed factor, and the other projects its

unrestricted factor into a subalgebra. The bounded self-adjoint parts of these algebras can serve as ideal seed families. Their invariance under this noise family, the full modular group and forward time translations makes their Γ_3 hulls exactly the displayed cone algebras.

Here $U(t) = e^{it(H_1+H_2)}$ is the physical time translation, $V_+ + te_0 \subset V_+$ for $t \geq 0$, and its only zero-energy vector is the vacuum. The latter follows from strictly positive one-particle energy $|p|$ almost everywhere. Gauge-fixed algebras are represented on their own vacuum-cyclic subspaces, not asserted cyclic on the ambient Fock space. Haar projection commutes with H; its restriction to each such subspace retains positivity, the same physical translation and the unique vacuum.

A compact neutral quadratic preparation in field 1, as specified in Consciousness II, lies in both readers' algebras, has strictly positive finite H energy and a smooth nonzero-frequency Radon source [6, Proposition on a neutral preparation]. Their selected algebras at every diamond contain the bounded stress and collar readings. Haar expectations preserve the common vacuum; permitted preparations, calibrated collar limits and common MC_η normalization remain inputs. The constraint-compatible linear response of that paper then instantiates Proposition 4.4. The reference hull construction uses the cone algebra; the geometric test uses the same gauge-fixed nets on all $D(B)$. This supplies the required diamond readings, not a claim that each diamond algebra is the cone's Γ_3 hull or that every ball operator belongs to $\mathcal{A}(V_+)$. The generic bounded pulse also exists here, but its source regularity is not inferred solely from its energy moments.

5 Agreement, calibration and fault transfer

A finite two-observer experiment tests the distinct roles of agreement and paid references. A bank of 4096 bits evolves under a hidden event $c_k = \pm 1$, flipping with probability $.01$ per epoch. Each bit is retained with probability $.65$, replaced by a fair coin with probability $.25$, or drawn with mean $.6c_k$ with probability $.10$. Each drifting binary instrument has mean response $g_i z + b_i$ and a 512-read budget per epoch. Known ± 1 references displace world reads; agreement exchanges current estimates and changes next-epoch bias calibration. Only the scorer accesses c_k. Appendix F and the supplementary protocol give the legal clipping, seeds, updates and full outputs.

For binary reports put $A = P(\hat{c}_1 = \hat{c}_2)$, $C_i = P(\hat{c}_i = c)$ and $E = P(\hat{c}_1 = \hat{c}_2 \neq c)$. Eventwise counting, without independence, gives

$$C_1 + C_2 = 1 + A - 2E, \qquad P(\text{correct} \mid \text{agree}) = 1 - E/A \quad (A > 0).$$

Scheme	A	C_1	E	$1 - E/A$
Initial references only	.647	.753	.067	.896
Agreement only	.837	.789	.125	.851
64 references per sign, no coupling	.860	.908	.020	.977

Table 1: Selected comparisons under the same 512-read budget: 800 trajectories of 3000 epochs, excluding the first 200; correctness standard errors below $.002$. The supplementary protocol retains all tested budgets and fault controls.

Agreement rises while the reliability of an agreeing pair falls. Among tested reference counts $4, 16, 32, 64, 128$ per sign, 64 gives the highest correctness; this is not a global optimum. The full common-bias scan and unequal-reference controls are retained with the protocol.

A 100-epoch fault starts at a random epoch and adds a clipped $.30$ bias or sets one instrument's gain to zero. With 16 references per sign, unweighted agreement transfers zero-gain correctness losses $.121, .123$ to the undamaged observer. Weighting by the partner's reported gain reduces them to $.014, .016$; 64 references reduce them to $.003, .004$. A matched no-coupling observer has zero transfer. A separately written implementation gives equal-weight harm $.110$ and weighted harm $.0036$. Isolation at a scorer-known fault time is a diagnostic, not a learned deployment policy. References remove bias; a zero-gain reading requires an additional gain-sensitive decision.

Communication costs 22 bits per epoch for agreement and 44 for weighted agreement. All observers start with 4096 reference reads. The world is uncontrolled, the rules are designed, and only two readers and the stated faults are tested. These data support calibration and fault transfer in this protocol, not a consensus-to-geometric law.

6 What makes a shared world testable

The intersection fixes common observables, restrictions of one actual state fix common state values, and instruments fix joint record laws. Correct public reports additionally require a reference that separates the public target's indistinguishability classes. This can leave private frames and old histories different. The algebraic intersection and the public-reference criterion answer complementary questions.

The common-pulse construction joins selected access to an actual source test. Its physical translation, preparation, source regularity, stress and collar readouts, and geometric consistency must coexist. Example 4.5 supplies that algebra/source combination in a conformal matter model. It retains the supplied coupling and gravitational data; the null generator in the chiral example is a different physical quantity. A sourced Einstein tensor does not determine free radiation, a metric gauge, or experience.

We propose that the operational extent of a shared world is selected partly by the references its participants can jointly sustain. Source changes alter reference availability and price; changed reference risk alters the legal common task family; the resulting actions change the next source. Some common capabilities can fail before private stored histories disappear. The finite reference witness establishes calibration and old-history preservation. Foundation III supplies a finite semiclassical mean-action return. One maintained reference/source/readout device remains a further requirement, rather than a consequence of a strict hull inclusion.

Failure conditions are concrete. A shared hull can be scalar, a reference can be conjugated with the faulty instruments, and a zero-gain instrument can erase the information needed to correct it. An energy reference cannot remove internal errors commuting with energy. Paid controls, output entropy and reusable memory determine the resource price; algebra size alone does not. Foundation III, T-DOME and Brain Anchor address the distinct return, calibration and formation mechanisms. Independent-source state equality and a common projected memory kernel retain their explicit appendix conditions.

A Independent sources: compatibility and its limits

The same-process construction uses restrictions of one actual state. For independent sources let M have a cyclic separating reference vector Ω, reference state ω_0, faithful normal states ϖ_i, and normal UCP semigroups $\mathcal{E}_t^{(i)}$. Suppose each source supplies an individual algebra $A_i \subseteq M$,

state $\omega_i = \varpi_i|_{A_i}$, and realized foreground space $V_i \subseteq M_{\text{sa}}$. The independently chosen ϖ_i need not agree on $A_1 \cap A_2$.

For a candidate $C \subseteq A_1 \cap A_2$, the independent-source conditions are:

JS-A. C is invariant under both semigroups and under the full reference modular group. The semigroups need not commute.

JS-B. A faithful normal state ψ on C is stationary under both restricted semigroups and satisfies $\omega_1|_C = \omega_2|_C = \psi$. Equality and stationarity are substantive inputs.

JS-C. When a joint projected record process is needed, its normal UCP idempotents P_1, P_2 commute; their product has von Neumann-algebra range containing C, fixes C, and the projected dynamics has one specified common predual memory kernel. This last clause is a dynamical condition, rather than a consequence of commuting projections [14, 15].

Proposition A.1 (Commuting expectations)**.** *If $E_i : M \to M_i$ are commuting normal conditional expectations, E_1E_2 is a normal conditional expectation onto $M_1 \cap M_2$. It supplies the algebraic clauses of JS-C when $C \subseteq M_1 \cap M_2$.*

Proof. The product is normal UCP and idempotent. Its range lies in both M_i, and every element of their intersection is fixed by it. The norm-one projection onto that subalgebra is a conditional expectation [18, 19]. No kernel identity follows. □

B An independent-source hull theorem

The state equality is an input in JS-B. The following conclusion constructs the minimal compatible hull inside a supplied candidate C; it does not derive agreement of independent states.

For a nonzero finite-dimensional $V_1 \cap V_2$, choose a basis $O_1, \ldots, O_k$ in C, and set $S = W^*(1, O_1, \ldots, O_k)$. The seed-space dimension is $k = \dim(V_1 \cap V_2) \leq \min(\dim V_1, \dim V_2)$. This bounds neither the algebraic dimension of S nor the size of its hull. A single observable $O = \operatorname{diag}(0, 1, 2)$ in M_3 spans a one-dimensional seed space but generates the three-dimensional diagonal algebra. One observable with infinite spectrum can likewise generate an infinite-dimensional algebra. The finite seed bound also says nothing about nontriviality. Let

$$\Phi = \bigcap \{N \subseteq M : N \text{ is a von Neumann algebra}, S \subseteq N,\ \sigma_s^{\omega_0}(N) = N,\ \mathcal{E}_t^{(i)}(N) \subseteq N\}.$$

Theorem B.1 (Independent-source compatibility)**.** *If C satisfies JS-A and JS-B and contains S, then*

$$S \subseteq \Phi \subseteq C \subseteq A_1 \cap A_2, \qquad \omega_1|_\Phi = \omega_2|_\Phi = \psi|_\Phi.$$

The state is faithful, and Φ inherits the reference modular flow on $\overline{\Phi\Omega}$. If a proper half-sided modular inclusion $N \subset \Phi$ is supplied there, with Ω cyclic and separating for N, Consciousness I's Cut correspondence applies. Standardness gives the net correspondence. A joint projected record process additionally requires JS-C.

Proof. JS-A makes C a competitor in the defining intersection, so the class is nonempty and $\Phi \subseteq C$. Intersections preserve the algebra, the seed inclusion and all specified invariances, proving minimality. Restriction preserves faithfulness, and JS-B gives equality of the states. Takesaki's theorem identifies the restricted reference modular group [17, 16]. The Cut and the projected kernel are the separate supplied inputs just stated. □

The Cut sits inside Φ in its reference representation; the hull is not asserted to be a Cut of the ambient algebra. The reference state $\omega_0|_\Phi$ need not equal $\psi|_\Phi$. The theorem does not turn agreement into an output when JS-B already assumes it.

Proposition B.2 (Finite compatibility witness)**.** *JS-A/B/C can coexist with disagreement elsewhere in the overlap.*

Proof. In $M_2(S) \otimes M_2(P_1) \otimes M_2(P_2)$ with tracial reference, let $D_X(a) = (a + Z_X a Z_X)/2$, $P_1 = D_S D_{P_1}$, $P_2 = D_S D_{P_2}$, and $\mathcal{E}_t^{(i)} = e^{-\lambda_i t}\mathrm{id} + (1 - e^{-\lambda_i t})P_i$, $\lambda_i > 0$. Choose $C = W^*(Z_S)$, $A_1 = A_2 = W^*(Z_S, Z_{P_1}, Z_{P_2})$, and

$$\varpi_1 = \rho(1/3)\otimes\rho(1/4)\otimes\rho(1/5), \quad \varpi_2 = \rho(1/3)\otimes\rho(-1/4)\otimes\rho(-1/5), \qquad \rho(a) = (1+aZ)/2.$$

All states are faithful. Both semigroups fix C pointwise, the reference flow is trivial, and the states agree on C, verifying JS-A/B. The projections commute and their product has diagonal range. On the fixed sector the common kernel is zero, verifying JS-C. The foregrounds $\mathrm{span}(Z_S, Z_{P_1})$ and $\mathrm{span}(Z_S, Z_{P_2})$ overlap in Z_S, while the state discrepancy on Z_{P_1} is $1/2$. This finite witness does not realize a Type III Cut. □

C Private content and common modular flow

A private complement $P_i^{\mathrm{priv}} \subseteq A_i$, when one is used, means $A_i = W^*(C \cup P_i^{\mathrm{priv}})$; no canonical tensor or free factorization is implied. A state-preserving expectation $E_i : A_i \to C$ is a separate regime condition.

Localization of discrepancy. For $\delta = \omega_1|_{A_1 \cap A_2} - \omega_2|_{A_1 \cap A_2}$, JS-B directly gives $C \subseteq \ker \delta$, so $\delta(x) \neq 0$ implies $x \notin C$.

The kernel need not be an algebra. In M_3, take states with densities $\rho = \mathrm{diag}(.5, .3, .2)$, $\sigma = \mathrm{diag}(.3, .4, .3)$, and $X = \mathrm{diag}(0, 1, -1)$. Then $(\rho - \sigma)(X) = 0$ but $(\rho - \sigma)(X^2) = -.2$. Nor do matching marginals specify a joint-record agreement rate.

Corollary C.1 (Common modular flow)**.** *If faithful normal $E_i : A_i \to C$ preserve ω_i, then*

$$\sigma_t^{\omega_1}|_C = \sigma_t^{\psi} = \sigma_t^{\omega_2}|_C.$$

It equals the reference flow used for a Cut if $\psi = \omega_0|_C$; equality of flows need not imply equality of states. For example, on $C = \mathbb{C}^2$, the faithful states with weights $(1/3, 2/3)$ and $(2/3, 1/3)$ differ but both modular flows are trivial. No factor condition is assumed here.

Proof. Apply Takesaki's theorem to each expectation and its preserved state. □

Example C.2 (A Type III shared sector with different source states)**.** Let $B = \mathrm{Vir}_1(0, \infty)$, with vacuum state ω_B; let $K, P_1, P_2 = M_2$ with normalized trace τ. Put $M = B\bar{\otimes}K\bar{\otimes}P_1\bar{\otimes}P_2$, $C = B \otimes 1$, and

$$A_1 = B\bar{\otimes}K\bar{\otimes}P_1 \otimes 1, \qquad A_2 = B\bar{\otimes}K \otimes 1\bar{\otimes}P_2.$$

With faithful η_i and $\rho_\pm = (1 \pm aZ)/2$, $0 < a < 1$, choose $\varpi_1 = \omega_B \otimes \rho_+ \otimes \eta_1 \otimes \tau$ and $\varpi_2 = \omega_B \otimes \rho_- \otimes \tau \otimes \eta_2$. They agree on C, differ by $2a$ on Z_K, and their partial state expectations onto C preserve the respective states. The tracial finite-factor reference makes the reference flow on C the vacuum dilation flow. Strong additivity of the $c = 1$ Virasoro net makes $\mathrm{Vir}_1(b, \infty) \subset B$, $b > 0$, a standard Cut on its own vacuum space [1, 5]. This witnesses the algebra, state and Cut conditions together; source semigroups and formation of foregrounds have not been specified.

D Finite families of independent sources

Theorem B.1 extends to any finite number of sources by requiring one candidate $C \subseteq \bigcap_i A_i$, one shared seed algebra in C, JS-A/B for every source, and one Cut inside the resulting hull. Its proof is the same intersection and restriction argument. If joint projected records are required, pairwise commuting normal UCP idempotents give an idempotent finite product; algebraic range and one common kernel still need to be supplied. Pairwise choices of overlap sectors and kernels do not establish one coherent joint process. The same-source theorem in the main text already gives the largest common admissible algebra without these independent-state conditions.

E Imported interfaces

Foundation I supplies the hull construction, stationary fixed algebras, restricted modular groups and conditional Cut inheritance [1]. Its projected stationary fixed-sector memory kernel vanishes; a general projected kernel remains extra dynamical data. Consciousness I supplies the half-sided modular inclusion/net correspondence, and Consciousness II supplies the bounded pulse and conditional higher-dimensional identification used in Section 4 [5, 6]. T-DOME's record value, history-selected frame and priced calibration are different functional interfaces [11, 12, 13]. They do not derive JS-B or preparation permission from a fixed algebra. The no-foreground-overlap example and the independent-state witnesses answer different questions and cannot replace one another.

F Simulation protocol for Section 5

The complete protocol is retained in the supplementary source record `FII_protocol.tex`, with the measured tables. Its implementation is `d2_v3.py`; the selected independent fault checks use `d2_checks.py`. The source map identifies the fixed scripts and outputs. All reads use legal clipped Bernoulli instruments; fault and no-fault runs share the world and per-reader random streams. Calibration uses current reference reads before the current report, while agreement updates the following epoch. Communication is quantized to eleven bits per scalar. The scorer's fault-time isolation is kept separate from deployable detection.

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